



Trueseal

Forensic Certificate Verification

Squad Hackathon 3.0 · Challenge 01: Proof of Life

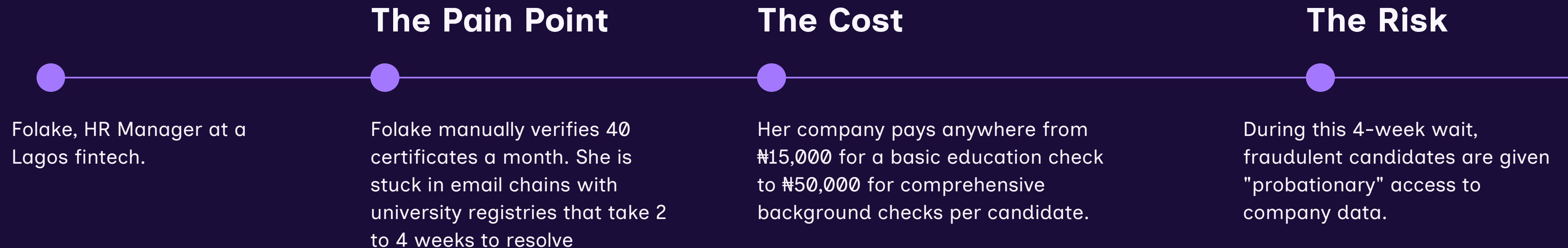
The Problem

- Certificate fraud is a systemic crisis threatening Nigerian employers.
- An estimated 33% of Nigerian graduates use fake credentials to secure employment.
- 68% of organizations encountered at least one major fraud attempt recently.
- The National Universities Commission (NUC) has currently blacklisted 58 active fake universities (degree mills) operating in Nigeria.



The Target

HR Teams are trapped in a costly, low-trust verification loop



Solution Overview

TrueSeal is a B2B forensic verification platform that completely automates document trust while creating a brand-new passive revenue stream for academic institutions.

Upload

HR uploads certificate and pays ₦5,000 via Squad Checkout



Scan

AI runs ELA heatmap + OCR forensic scan on the document



Pay Out

₦1,000 royalty auto-transferred to university bank account



Verdict

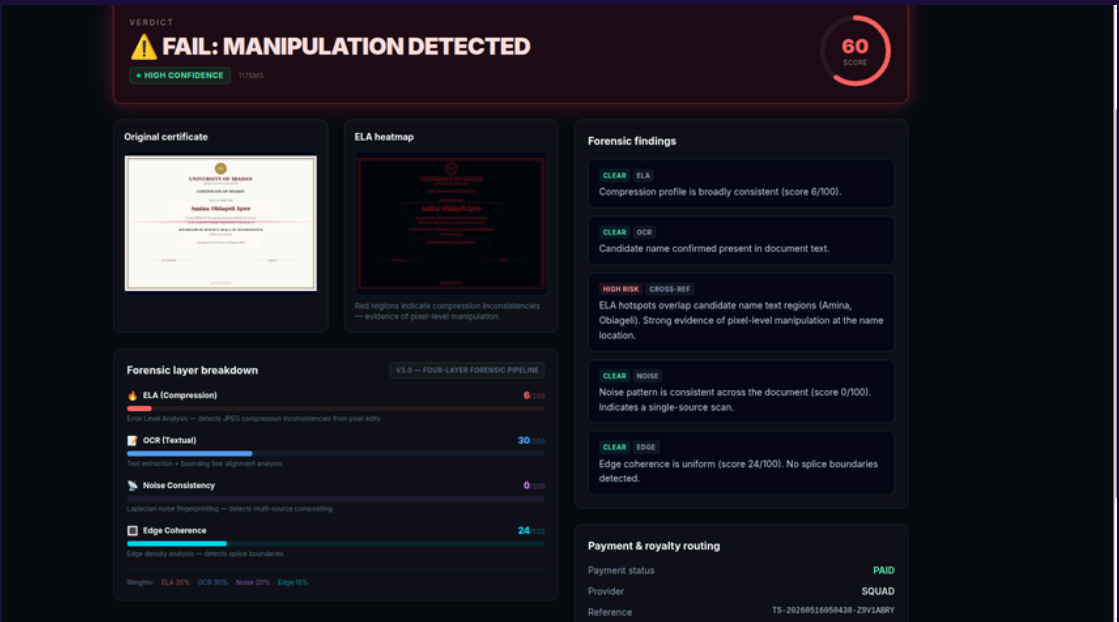
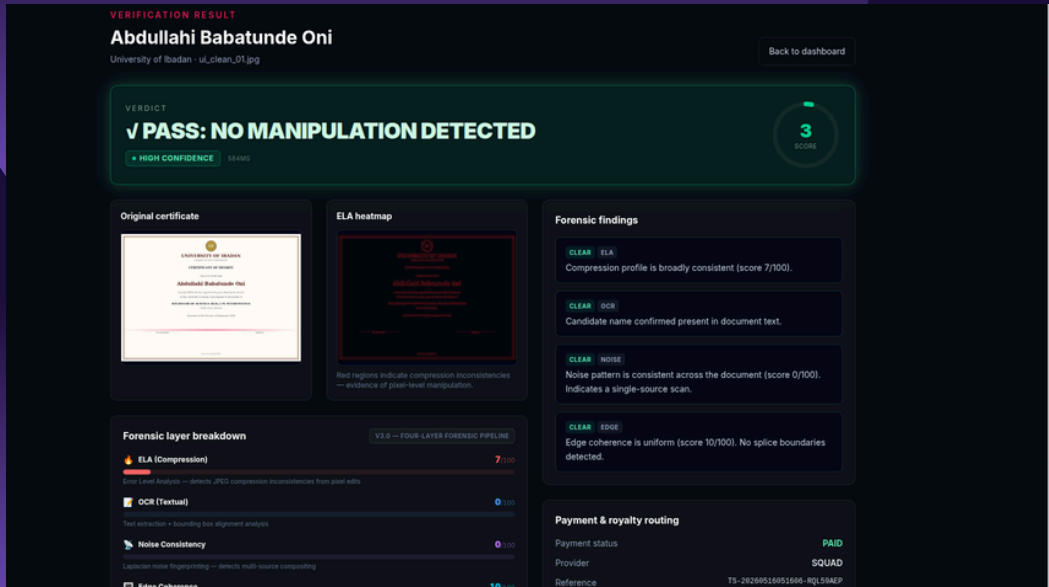
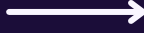
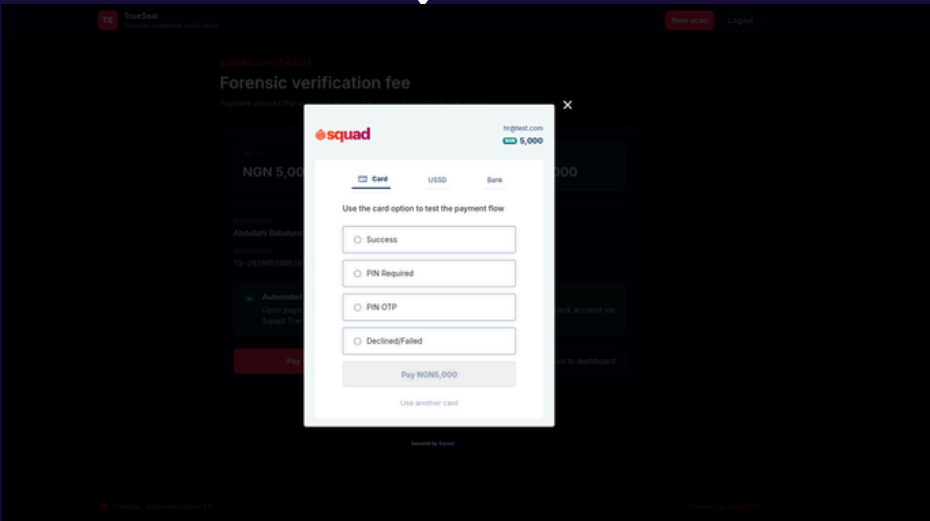
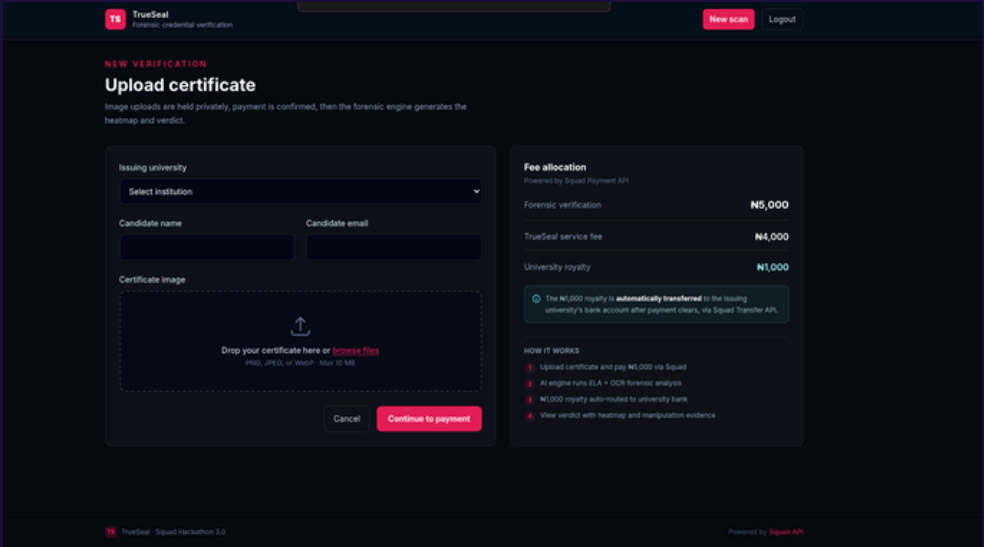
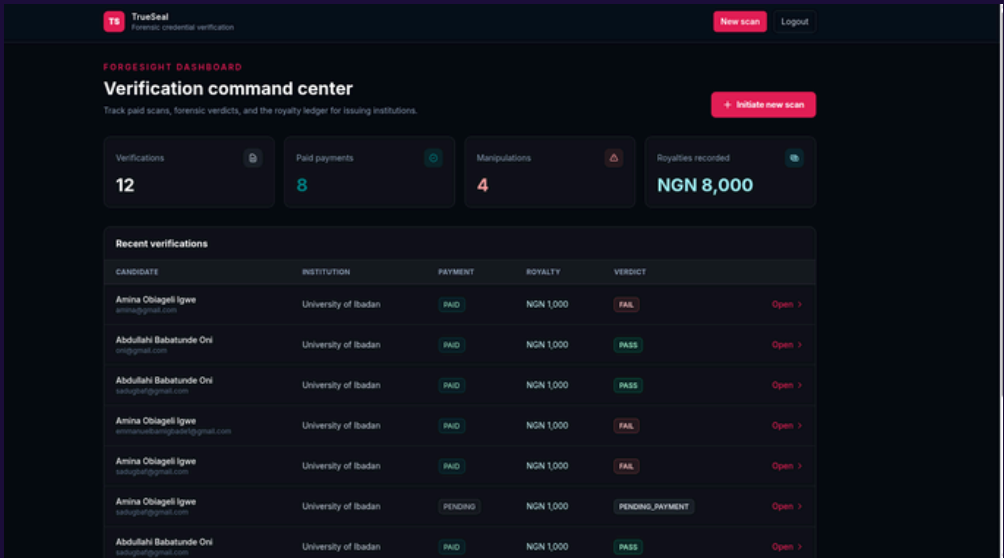
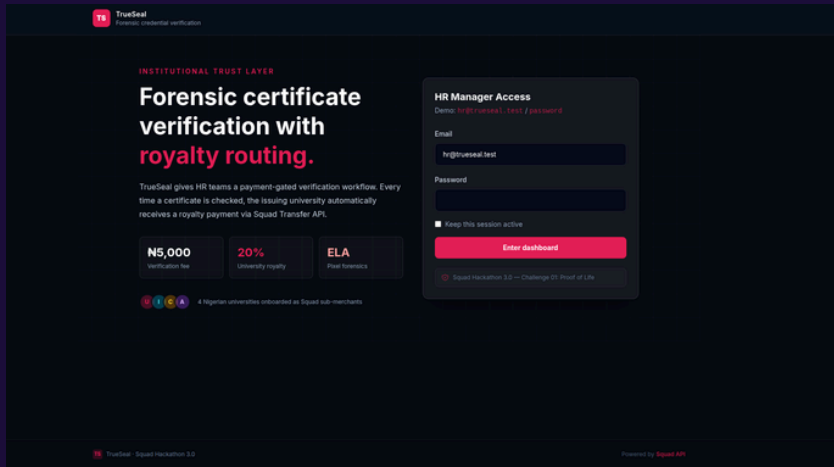
Verdict rendered: PASS or FAIL with visual evidence report

Squad API Integration

TrueSeal does not just analyze files; it acts as an automated clearinghouse powered by a 4-endpoint Squad API integration architecture.

API	PURPOSE	ENDPOINT
Initiate Payment	Collect ₦5,000 verification fee via inline modal	POST /transaction/initiate
Verify Transactions	Server-side confirmation before releasing results	GET /transaction/verify/{ref}
Transfer API	Route ₦1,000 royalty to university bank account	POST /payout/transfer
Sub-Merchant	Register universities as aggregator sub-merchants	POST /merchant/create-sub-users

User Flow



AI / Data Intelligence

TrueSeal moves beyond basic database lookups. If a document has been digitally altered, its pixel structure will expose it. We process documents through a weighted, Four-Layer Forensic AI Engine.

ELA (Error Level Analysis) detects compression inconsistencies from pixel edits. Forged names and dates show as red hotspots on the heatmap.

Visual Layer
Weight: 35%

OCR via Tesseract 5.3 verifies candidate name presence, detects bounding box alignment anomalies, and cross-references ELA regions.

Textual Layer
Weight: 30%

Noise Consistency derived using Laplacian filters to detect deviations in noise variance, catching edits sourced from editing tools.

Noise Consistency
Weight: 20%

Edge Coherence Employs Sobel operators to scan localized edge-density variations around character boundaries, highlighting cropped and pasted text regions.

Edge Coherence
Weight: 15%

Impact Potential Scalability & Business Model

Total Addressable Market

Nigeria's higher education system produces roughly 500,000 graduates annually across its universities, polytechnics, and colleges. At our target entry fee of ₦5,000 per check, the baseline immediate verification addressable market for fresh graduates alone is ₦2.5 Billion annually.

Serviceable Addressable Market

Industry reports indicate that modern verification and screening adoption within structured corporate sectors sits around 46%. This concentrates our immediate addressable market to a highly reachable segment of ₦1.15 Billion.

The Lifetime Backlog

The market scales exponentially when factoring in historical backlogs. Ten years of historical graduate certificates represent a massive pool of over 5 Million active documents within the workforce, blowing the total potential verification market past ₦25 Billion.

Revenue Architecture

- The Core Split: Out of every ₦5,000 verification scan, ₦4,000 is retained as TrueSeal platform revenue, while ₦1,000 is instantly routed as a royalty payment back to the issuing university via Squad's Transfer API.
- The University Advantage: Zero setup costs, zero technological overhead, and zero continuous effort for schools. Academic institutions transition from viewing verification as an administrative bottleneck to embracing it as an automated, recurring revenue stream.

Future Product Roadmap

- Phase 1 (Current): Pixel-level ELA, OCR boundary matching, and automated sub-merchant royalty distribution.
- Phase 2 (Scale): Introducing Structural Template Matching. The platform will map incoming documents against official layouts, stamp positions, and cryptographic signature boundaries unique to each partner university to block advanced, layout-accurate visual recreations. Our Partnership with universities will make them share the necessary training data with us

Research & Validation

The Daily Nigerian Degree Mill Exposé (2024): An undercover investigative report revealed a cross-border syndicate issuing fully verified university degrees from the Benin Republic in under six weeks, completely bypassing academic structures. This incident forced the Federal Ministry of Education to temporarily halt the evaluation of degrees from multiple foreign nations, creating an immediate verification backlog for thousands of cross-border professionals entering the local workforce.

Discussions with corporate HR managers highlight a clear consensus: teams are completely willing to pay a premium to eliminate the weeks-long delays and human errors of manual verification, provided the alternative offers absolute legal security and verifiable accuracy.

The Mmesoma Ejikeme Case (2023): A national-scale incident where a UTME candidate manually manipulated her result printout using a mobile application to artificially boost her score. The forgery was only exposed after manual intervention by JAMB's technical team, highlighting the urgent need for instant, automated pixel-level verification.

The Team

Meet the Team: A two-man team combining legal domain expertise with full-stack execution).



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Software Engineer & Law Student
(AI & Backend Architecture)



Adegboye Habeebullah Temitope
UI/UX Designer & Law Student
(Product Design & Frontend UX)